First- versus second-order mimetic velocity reconstruction in `wrf_ftle`: where the FTLE differs and why

pv_ftle — branch `2ndorder`

November 1, 2026

Abstract

The first-order (lowest-order Raviart–Thomas) and second-order ($Q_{2,1,1}$) velocity reconstructions of `wrf_ftle` were compared on WRF output over the Transantarctic Mountains and the Ross Ice Shelf (2.3×10^6 seeds, $T = -1\,\mathrm{h}$). Over most of the domain the two FTLE fields agree closely (median difference 2% of the FTLE scale) and the differences are unbiased, largest along sharp FTLE ridges. The exception is a predictable, systematic first-order artefact: in cells into which the (time-reversed) vertical flow converges, both seeds bounding the cell are drawn into the same layer, where the first-order wind is constant with height, so they do not separate and the first-order FTLE collapses. This affects the ground layer most (first-order FTLE 25% lower on average, holes in 15% of its cells) and the thick upper layers. The differences between the orders are 100–1000 times larger than the time-stepping error, so they come from the reconstruction. Coarsening the data by 2 and 4 shows the difference between the orders shrinking roughly in proportion to the grid spacing, and the second-order FTLE changing about three times less than the first-order one under coarsening: the evidence points to both converging to the same field, with the first-order error dominating at WRF's native 3.3 km resolution. An independent integration with NumbaCS-3D (cubic splines, adaptive DOP853) agrees with the second-order result two to three times better than with the first-order one, and shows none of the first-order holes.

1 Introduction

`wrf_ftle` advects particles in index space $(p, q, r) = (i + \xi,\, j + \eta,\, k + \zeta)$ with a mimetic velocity reconstruction built from the WRF face fluxes, and computes the finite-time Lyapunov exponent (FTLE) from the flow map. Two reconstructions are available (`--order 1|2`). This report compares them on WRF output over the Ross Ice Shelf and the Transantarctic Mountains, asking three questions:

1. Where do first- and second-order FTLE fields differ?
2. Are the differences predictable from properties of the flow or grid?
3. Do the two converge to the same FTLE field as the grid is refined?

2 The two reconstructions

In a cell, the velocity in index space is $\dot{\xi} = F_\xi / J$ (likewise for η, ζ), where J is the Jacobian of the trilinear map and F_ξ the flux density through surfaces of constant ξ. The face data are the face-mean normal velocity $w = \phi / A$ (flux over area) and the face area A.

Order 1 (lowest-order Raviart–Thomas): F_ξ varies linearly between the two ξ-faces of the cell and is constant over each face,

$$F_\xi = (1 - \xi)\, A_0 w_0 + \xi\, A_1 w_1. \tag{1}$$

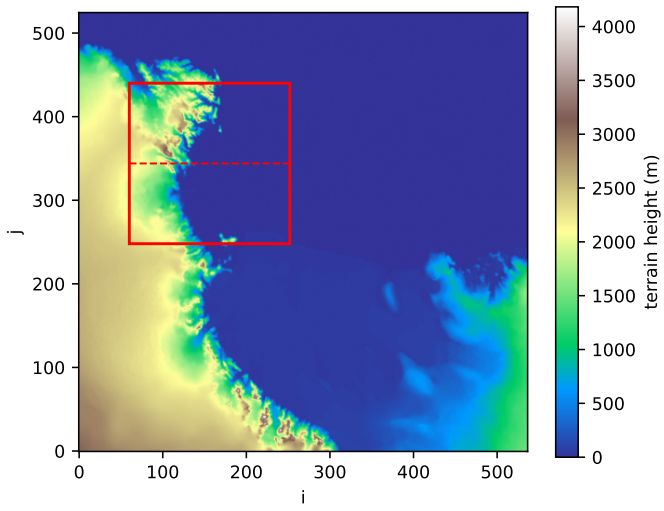


Figure 1: Terrain height of the WRF domain, the study window (red box) and the cross-section of Fig. 4 (dashed).

The normal component is continuous across faces, but the tangential components are piecewise constant: they *jump* across every face. In particular, the horizontal wind is constant with height inside each layer.

Order 2 ($Q_{2,1,1}$): F_ξ is quadratic between the faces and linear along each face,

$$F_\xi = (1-\xi)\,A_0\big[w_0+s_\eta^{(0)}(\eta-\tfrac{1}{2})+s_\zeta^{(0)}(\zeta-\tfrac{1}{2})\big]+\xi\,A_1\big[w_1+s_\eta^{(1)}(\eta-\tfrac{1}{2})+s_\zeta^{(1)}(\zeta-\tfrac{1}{2})\big]+\tfrac{1}{2}\bar{A}\,c\,\xi(\xi-1), \quad (2)$$

where the tangential slopes s are central differences of w over the neighbouring faces (one-sided at the domain edge) and c is the mean second difference of w across the cell's two faces. The slope terms integrate to zero over each face and the quadratic term vanishes on both faces, so every face carries exactly its WRF flux, and the ground and model lid remain impermeable. Differencing w rather than ϕ keeps a constant wind exact on stretched layers. On box grids, order 2 reproduces a linear wind to round-off and makes all velocity components continuous across faces (`tests/test_wrf_reconstruction.py`). It costs about 2.3× the first-order RK4 time.

3 Data and method

Data. WRF output `wrf_f005.nc` (polar stereographic, $\Delta x = 3.33$ km, 60 terrain-following layers, lowest layer ≈ 25 m thick), with `wrf_f004.nc` for the time-dependent case.

Study window. 192×192 columns, $i = 60$–252, $j = 248$–440 (Fig. 1): East Antarctic plateau, the Transantarctic Mountains (slopes above 3% in 15% of the window) and the flat Ross Ice Shelf (53% of the window below 10 m). Seeds are placed at every cell corner of the window on all 61 levels (2.27×10^6 seeds); particles move over the whole domain.

Integration. Backward in time, $T = -3600$ s, frozen wind from `wrf_f005` unless stated otherwise, RK4 with CFL 0.5 (509 steps), using the C++ integrator (double precision). FTLE is the cell-centred 3-D FTLE of `wrf_ftle`. Differences are reported relative to the FTLE scale $S = 1.40 \times 10^{-3}\,\mathrm{s}^{-1}$, the 99th percentile of the order-2 FTLE. Approximate heights above ground of the levels quoted below: level 0 13 m, 1 45 m, 3 145 m, 5 290 m, 10 880 m, 20 3.1 km, 30 7.0 km, 40 13.6 km, 55 27 km.

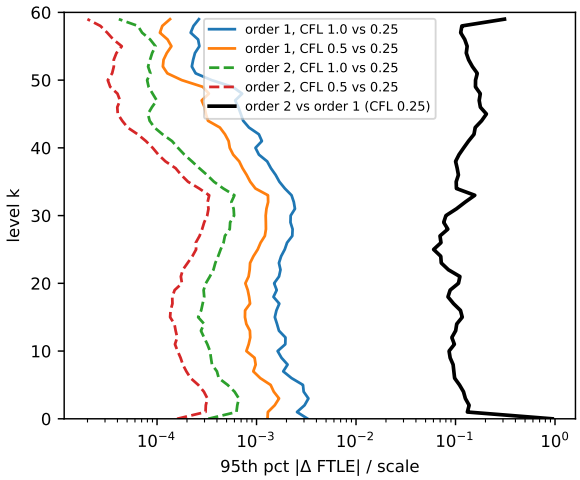


Figure 2: 95th percentile, per level, of the FTLE change between CFL 1 or 0.5 and CFL 0.25 for each order, against the difference between the orders (black).

Separating the effects. The two orders are compared with the same seeds, time step and grid, so any difference comes from the reconstruction. A time-step series (CFL 1, 0.5, 0.25) checks that time-stepping error is small compared with the order difference.

Grid refinement. WRF cannot be refined, so the data were *coarsened* by $c = 2$ and 4: every c-th corner is kept and fine face fluxes are summed over $c \times c$ sub-faces (with their areas). This conserves the flux through every coarse face exactly (checked to 10^{-15}), so the coarse grids are nested, flux-consistent versions of the fine one. A common set of seeds (every 4th corner and level, $49 \times 49 \times 16$) is tracked on all three grids with both orders and the same time step, and the final positions and FTLE are compared. If both orders converge to the same limit, their difference must shrink as the grid is refined.

4 Results

4.1 Overall size of the differences

Over all 2.2×10^6 cells, $|\mathrm{FTLE}_2 - \mathrm{FTLE}_1|/S$ has a median of 2.2%, a 95th percentile of 15% and a 99th percentile of 33%; 72% of cells differ by more than 1% and 10% by more than 10%.

The time step is not the cause (Fig. 2). Halving CFL from 0.5 to 0.25 changes the FTLE by a median of $4.0 \times 10^{-5}S$ at order 1 and $6.6 \times 10^{-6}S$ at order 2 (95th percentiles $7.6 \times 10^{-4}S$ and $1.5 \times 10^{-4}S$), while the two orders differ by a median of $2.1 \times 10^{-2}S$ at the same CFL. The order difference is therefore a spatial-discretisation effect, 100–1000 times larger than the time-stepping error; order 2 is also about five times less sensitive to the time step, consistent with its smoother velocity field.

4.2 Where the differences occur

Figures 3–5 show three regimes.

The ground layer (level 0). The largest and only strongly systematic differences. The first-order field has sharp-edged "holes" of near-zero FTLE, mostly over the flat ice shelf (Fig. 3, top row, and the near-surface blocks in Fig. 4) that are absent at order 2. The median difference is 10.5% of S and the 95th percentile 99%; the mean first-order FTLE of the layer is 25% lower (1.02×10^{-3} versus $1.36 \times 10^{-3}\,\mathrm{s}^{-1}$, Fig. 5, right).

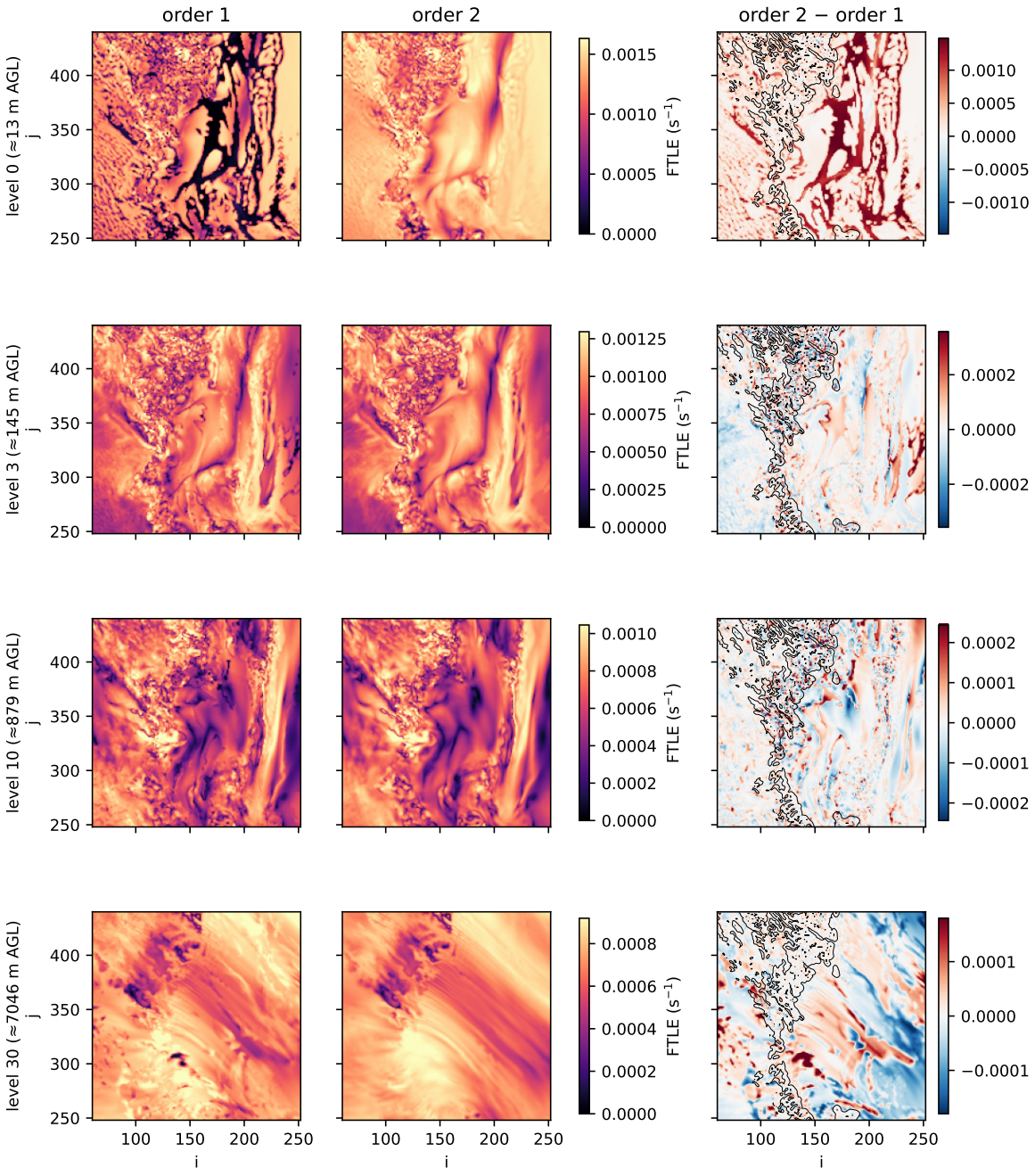


Figure 3: FTLE of order 1 (left) and order 2 (centre) and their difference (right) at four levels. Black contours on the difference maps: terrain slope 3%.

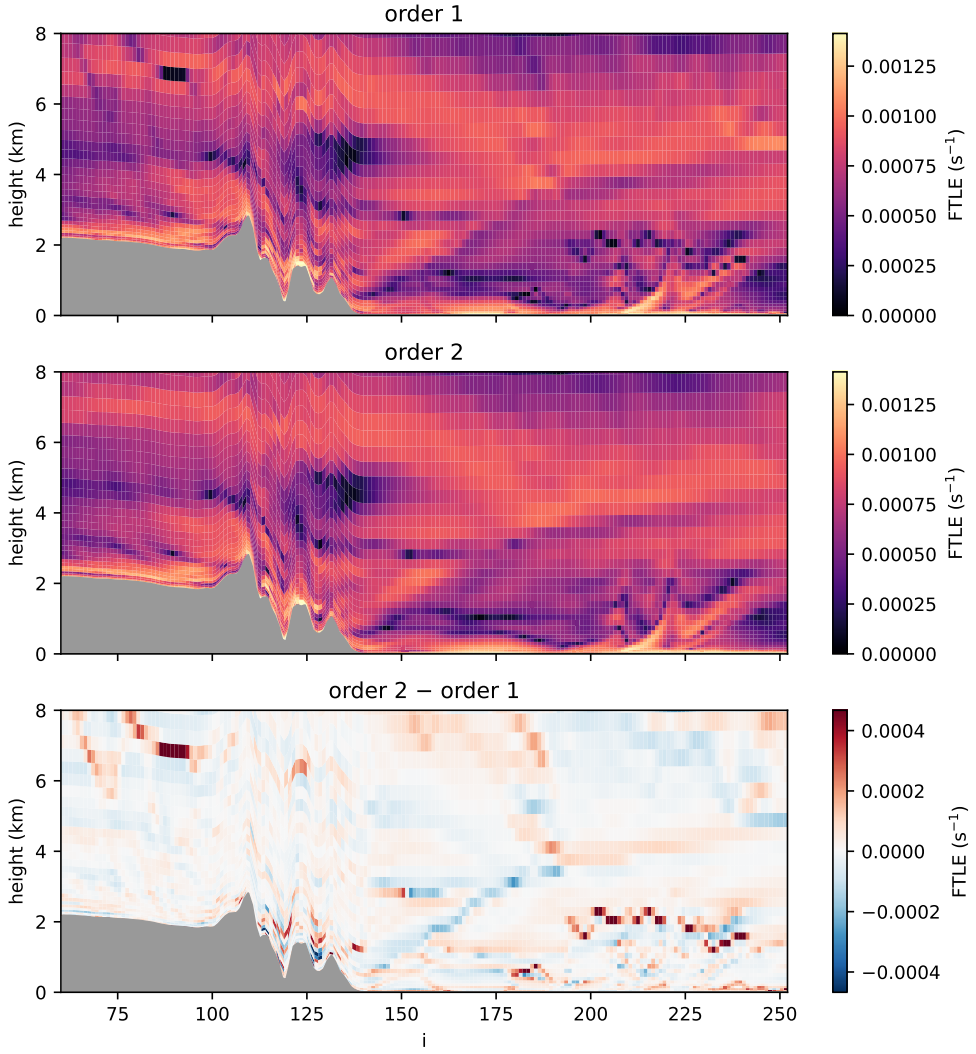


Figure 4: Cross-section along $j = 344$ from the plateau (left) over the mountains to the ice shelf (right), lowest 8 km. Grey: terrain.

The free atmosphere (levels 1–40). Small and unbiased: median differences 1.5–3.3% of S, mean FTLE per level equal to within about 1%. The differences are fine-scale and of both signs, lying along the edges of FTLE filaments (levels 3 and 10 in Fig. 3); aloft (level 30, 7 km) they become smooth, larger-scale patterns of about 10% of S, where the layers are several hundred metres thick and the first-order wind is constant over that depth.

The thick upper layers (levels $\gtrsim 40$, above 14 km). Median differences rise again to 4–6% and first-order holes reappear (Fig. 6, left).

4.3 A predictable pattern: trapped seeds

The holes have a simple cause. At order 1 the horizontal wind is constant with height inside a layer, and a seed lying exactly on a level is assigned to the layer above it. Integrating backward in time, a seed on a level moves down where $W > 0$ on that level and up where $W \leq 0$. In a cell with $W \leq 0$ on its bottom face and $W > 0$ on its top face (a layer into which the time-reversed vertical flow converges), the seed on the bottom face stays in the cell and the seed on the top face drifts down into it: both then move with the *same* first-order wind and do not separate vertically, so the first-order FTLE collapses. At order 2 the wind varies linearly with height inside the layer and the two seeds keep separating. At the ground $W = 0$, so the condition reduces to $W > 0$ on

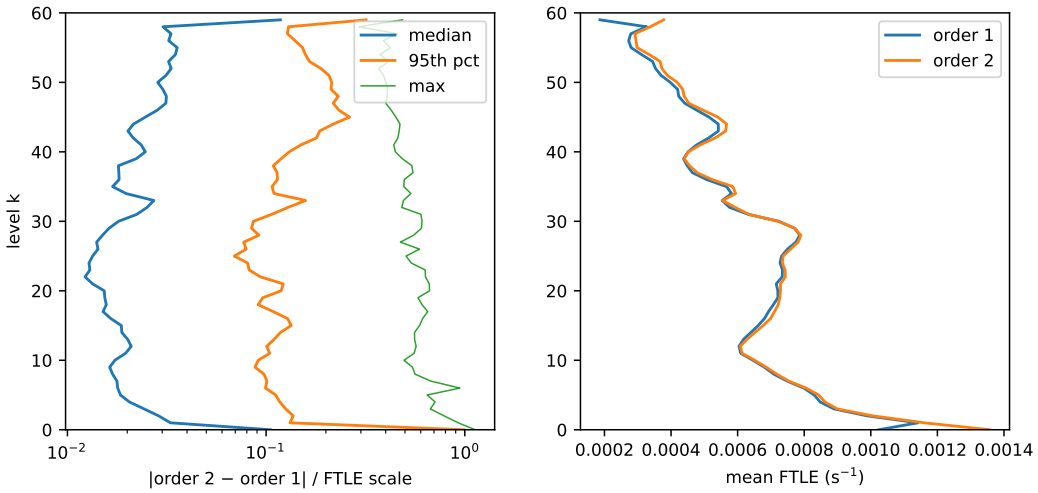


Figure 5: Left: distribution of $|\mathrm{FTLE}_2 - \mathrm{FTLE}_1|/S$ per level. Right: mean FTLE per level for both orders.

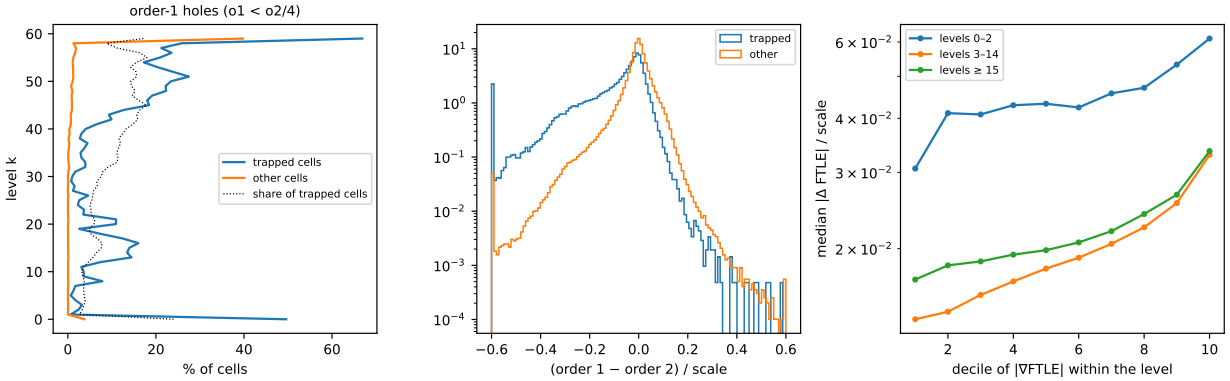


Figure 6: Left: percentage of cells with a first-order hole ($\mathrm{FTLE}_1 < \mathrm{FTLE}_2/4$) per level, for trapped cells ($W \le 0$ below, $W > 0$ above) and all other cells; dotted: share of trapped cells. Centre: distribution of $(\mathrm{FTLE}_1 - \mathrm{FTLE}_2)/S$ for trapped and other cells. Right: median $|\Delta\mathrm{FTLE}|/S$ against the decile of the FTLE gradient within each level.

level 1. For forward integration the signs flip ($W \ge 0$ below, $W < 0$ above).

Table 1 confirms the rule: trapped cells are 9.6% of all cells but contain 61% of the holes, with a hole rate of 15% against 1% elsewhere, and carry the systematic negative bias of order 1. The effect is strongest in the ground layer (half of its trapped cells are holes) and in the thick upper layers; in the thin layers of the lower and middle troposphere a trapped seed soon leaves the layer and the effect is weak (hole rates below 1% at levels 1–5). At level 0, the mean bias is −40% of S where $W > 0$ on level 1 and −11% where $W \le 0$.

The size of the wind jump between layers does *not* predict the differences (rank correlation −0.04 within trapped cells), so the effect depends on the vertical motion, not on the shear.

Away from trapped cells the differences are largest where the FTLE field is sharpest (Fig. 6, right): from the smoothest to the sharpest decile of the FTLE gradient the median difference rises from 1.4% to 3.3% (levels 3–14) and from 3.1% to 6.1% (levels 0–2), as expected if the reconstruction mainly shifts ridges slightly. Near the ground the differences also grow with terrain slope, from 4.2% (slope below 0.5%) to 6.2% (slope above 10%) of S in levels 0–2.

Time-dependent wind. With the wind interpolated between `wrf_f004` and `wrf_f005`, the statistics are almost identical (median 2.1%, 95th percentile 14%, level 0 median 10.9%; Fig. 7).

Table 1: First-order holes ($\mathrm{FTLE}_1 < \mathrm{FTLE}_2/4$) in trapped cells and in all other cells; F_1, F_2: first- and second-order FTLE.

	cells	hole rate	median F_1/F_2	mean $(F_1 - F_2)/S$
trapped cells	9.6%	15.2%	0.89	−10.1%
other cells	90.4%	1.0%	1.00	−0.5%
level 0, trapped	24%	50%	0.27	
level 55, trapped	18%	21%	0.63	

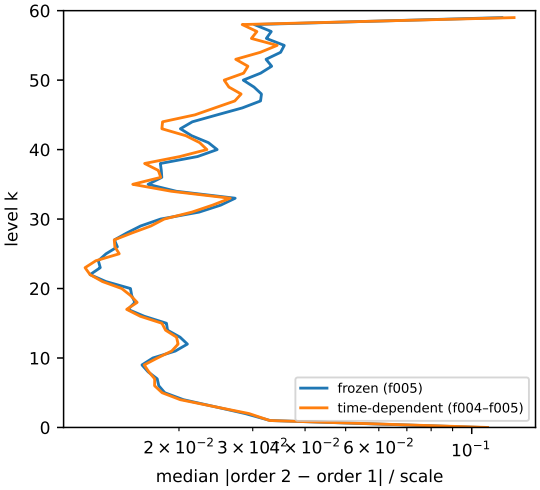


Figure 7: Median $|\mathrm{FTLE}_2 - \mathrm{FTLE}_1|/S$ per level, frozen versus time-dependent wind.

4.4 Do the two orders converge to the same field?

The difference between the orders shrinks steadily as the grid is refined (Fig. 8, Table 2). For final positions above the lowest seed levels it falls from 4.8 km ($c = 4$) to 2.3 km ($c = 2$) to 1.1 km ($c = 1$), about a halving per refinement (median displacement 26 km); for the FTLE it falls from 4.9% to 2.8% to 1.2% of S. Near the ground it shrinks more slowly (2.2 km, 1.5 km, 0.9 km), where the trapping artefact of order 1 depends on the vertical motion and not on the grid spacing alone.

Each order also converges towards its own finest-grid result. Second-order FTLE changes about three times less than first-order FTLE between $c = 2$ and $c = 1$ (0.7% versus 2.0%), and the second-order field at 6.7 km is closer to the second-order field at 3.3 km than the two orders are to each other at 3.3 km (0.7% versus 1.2%). Visually (Fig. 9), order 2 keeps the structure of the ground-layer FTLE down to $c = 4$, while order 1 develops large holes as the layers thicken.

With only three resolutions, and the native WRF grid as the finest, these figures are estimates of the trend, not a proof of convergence. They are, however, consistent with both reconstructions converging to the same FTLE field, with an inter-order difference that decreases roughly in proportion to the grid spacing — the signature of the first-order error dominating it.

5 Independent check with NumbaCS-3D

The comparisons above cannot say which order is closer to the truth. As an independent reference, the same seeds were integrated with the 3-D development version of NumbaCS (A. Jarvis, `flows_3D.py`, `integration_3D.py`), which interpolates a velocity given on a uniform grid with tensor-product cubic splines (or linearly) and integrates with the adaptive DOP853 scheme (relative tolerance 10^{-6}). Both the interpolation and the integrator are unrelated to `wrf_ftle`'s.

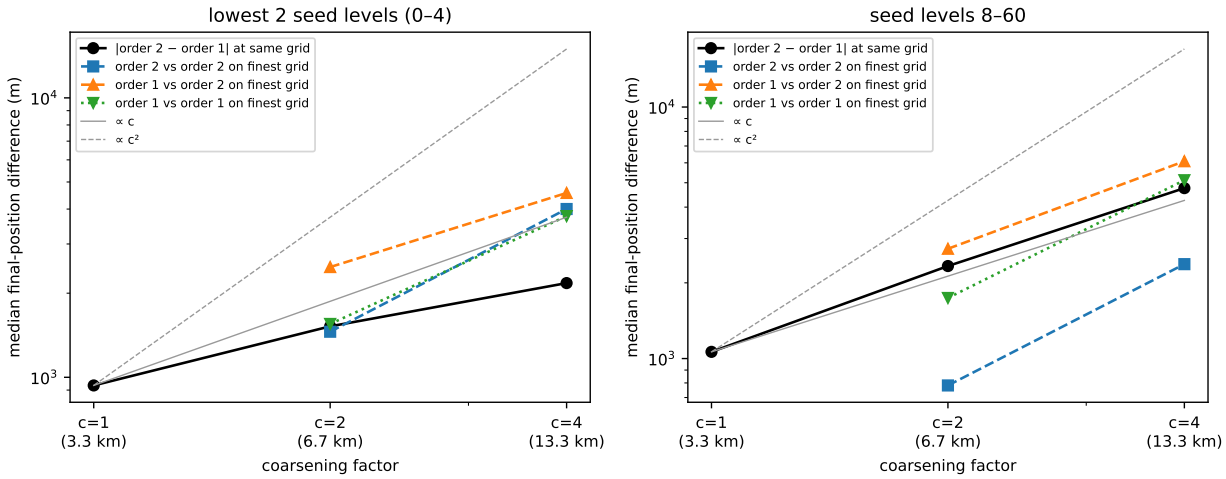


Figure 8: Median distance between final particle positions on the common seed lattice for grids coarsened by $c = 1, 2, 4$: between the orders on the same grid (black), and of each order against the finest grid. Left: the two lowest seed levels; right: seed levels 8–60. Grey: slopes $\propto c$ and $\propto c^2$.

Table 2: Median FTLE differences on the common seed lattice, relative to S, for grids coarsened by c (3.3 km, 6.7 km, 13.3 km).

	$c = 1$	$c = 2$	$c = 4$
order 2 vs order 1, same grid	1.2%	2.8%	4.9%
order 2 vs order 2 at $c = 1$	—	0.7%	3.0%
order 1 vs order 1 at $c = 1$	—	2.0%	5.8%

Setup. NumbaCS-3D needs a uniform grid, so it runs in the same index space (p, q, r). Its input is the usual destaggered velocity: horizontally at cell centres, vertically on the model levels. $\dot{p}, \dot{q}$ are the cell-centre values (mean of the two face fluxes over J), averaged between the layers below and above each level and extrapolated linearly to the ground and lid; $\dot{r}$ is the face flux over J on each level, hence exactly zero on the ground and the lid, which the tensor-product splines keep impermeable. The grid covers the study window plus a 50-cell margin (more than the farthest displacement, about 43 cells); no particle came within 3 cells of its edge. Seeds: every second corner of the window on all 61 levels (5.7×10^5 seeds); frozen wind from `wrf_f005`, $T = -3600$ s; `wrf_ftle` at CFL 0.25. Final positions of all methods are converted to metres with the same trilinear map and the FTLE is computed with the same code, so only the trajectories differ.

Results. The two NumbaCS interpolants agree with each other closely (median FTLE difference 0.16%), so they provide a consistent reference (Table 3). The second-order result is two to three times closer to it than the first-order one in FTLE (median 1.05% versus 2.4%, 99th percentile 11% versus 29%), three times closer in final positions above the ground and thirty times closer at the ground (60 m versus 1890 m). The first-order deviation from NumbaCS is nearly the same as its deviation from order 2, i.e. the difference between the orders is almost entirely first-order error. By level (Fig. 10) the first-order deviation peaks in the ground layer (median 11%) and the top layer (11%), where order 2 deviates by 1.1% and 0.4%.

NumbaCS shows none of the first-order holes: cells with FTLE $<$ FTLE$_{\text{cubic}}/4$ make up 1.5% of cells at order 1 (9.3% of the ground layer) against 0.04% at order 2 (0.01%) (Fig. 11). What remains between order 2 and NumbaCS is small-scale and of both signs, mostly along FTLE filaments over the mountains, where the two interpolations of the unresolved small scales differ.

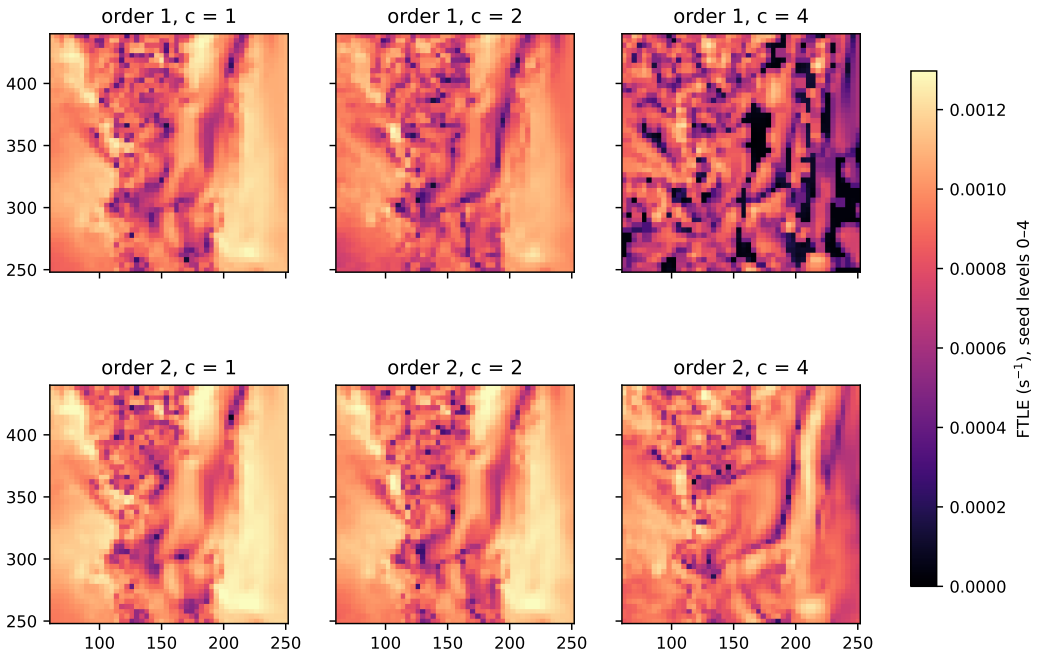


Figure 9: FTLE of the lowest cell layer of the common seed lattice (seed levels 0–4) on the native ($c = 1$) and coarsened ($c = 2, 4$) grids.

Table 3: Median distance between final positions (median displacements 10.6 km, 14.6 km and 27.4 km) and FTLE differences relative to the scale S (99th percentile of the NumbaCS cubic FTLE), 5.7×10^5 seeds.

	final position (m)			FTLE / S		
	ground	levels 1–10	levels 11–60	median	95th	99th
order 1 vs NumbaCS cubic	1890	499	1168	2.4%	14%	29%
order 2 vs NumbaCS cubic	60	242	404	1.05%	6.0%	11%
order 1 vs order 2	1904	486	1134	2.1%	14%	29%
NumbaCS cubic vs NumbaCS linear	43	92	56	0.16%	1.3%	3.2%

6 Conclusions

1. **Where.** Over most of the atmosphere the two orders give nearly the same FTLE: median differences of 1.5–3% of the FTLE scale, unbiased, concentrated along sharp FTLE ridges. Large, systematic differences occur in the ground layer (first-order FTLE 25% lower on average) and in the thick upper layers above about 14 km.

2. **Predictable pattern.** The systematic differences are a first-order artefact with a simple rule: in cells into which the (time-reversed) vertical flow converges — $W \leq 0$ on the bottom face and $W > 0$ on the top face for backward FTLE — both bounding seeds share one layer's constant first-order wind and the FTLE collapses. These cells hold 61% of the first-order holes. Order 2 avoids this because its wind varies linearly with height inside each layer.

3. **Convergence.** The difference between the orders shrinks roughly in proportion to the grid spacing under coarsening, and order 2 is about three times less sensitive to resolution, so the evidence is that both converge to the same FTLE field, with order 1's error dominating at WRF's native resolution. At 3.3 km the remaining median difference is about 1–2%, but individual first-order cells can be wrong by 100%.

4. **Independent check.** NumbaCS-3D (cubic splines, DOP853) agrees with order 2 two to three times better than with order 1 and has none of the first-order holes, so the second-order

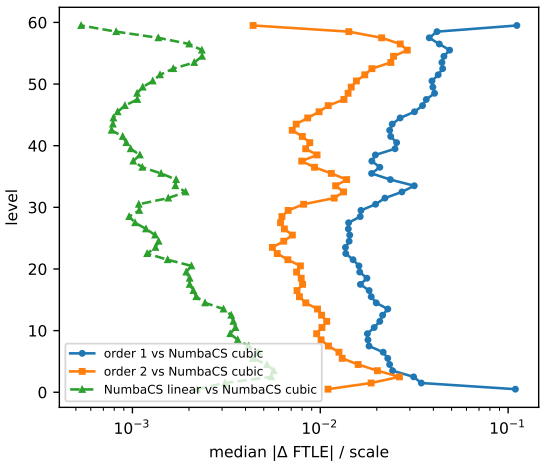


Figure 10: Median $|\Delta\mathrm{FTLE}|/S$ per cell layer of each method against NumbaCS cubic.

result is the more accurate one.

5. **Recommendation.** Use order 2 (the default). Treat first-order FTLE in the lowest layer and in thick upper layers with caution, especially where the vertical velocity changes sign across a layer. The extra cost of order 2 (about 2.3× the RK4 time) is small compared with the accuracy gained.

A Reproducing

```
python scripts/order_study_run.py WRF_DIR DATA_DIR
python scripts/order_study_figures.py DATA_DIR WRF_DIR \
       report/order2/figures
cd report/order2 && latexmk -pdf order2.tex
```

NumbaCS-3D comparison (stage 2 runs in the environment of the NumbaCS-3D development scripts, here `~/work/CSP-224-FTLE`):

```
STRIDE=2 python scripts/numbacs_compare_export.py WRF_DIR export.npz
NUMBACS3D/venv/bin/python scripts/numbacs_compare_run.py \
       NUMBACS3D export.npz numbacs.npz
python scripts/numbacs_compare_analyse.py WRF_DIR export.npz \
       numbacs.npz report/order2/figures
```

`WRF_DIR` holds `wrf_f004.nc` and `wrf_f005.nc`. Numbers quoted in the text are in `figures/stats.json` and `figures/numbacs_stats.json`.

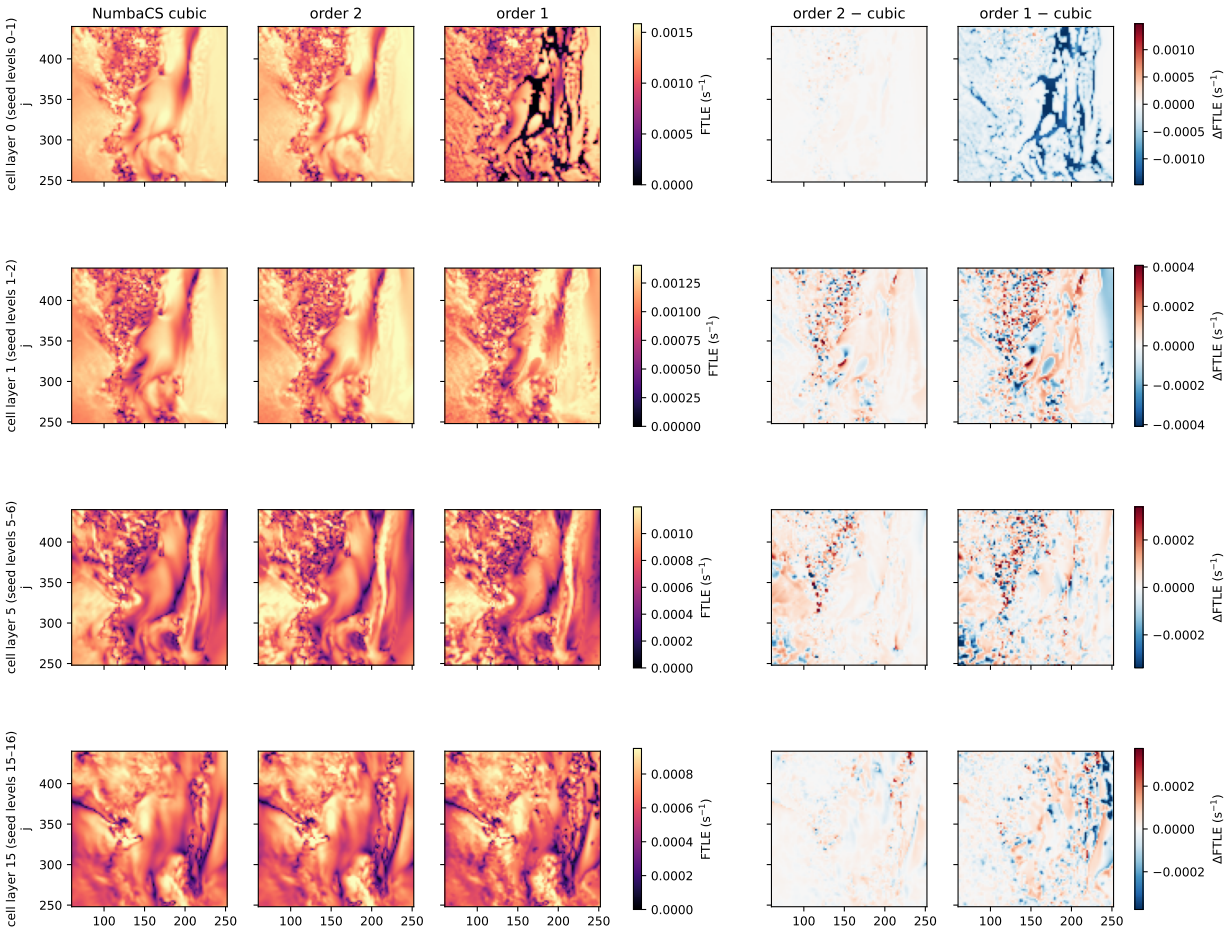


Figure 11: FTLE from NumbaCS-3D (cubic), `wrf_ftle` order 2 and order 1, and the differences of the two orders from NumbaCS, for four cell layers of the seed lattice.